

Oguz Kagan Hitit

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Education

University of California, Los Angeles, Los Angeles, CA Sep 2026 – Present
M.S. in Electrical & Computer Engineering

Koç University, College of Engineering, Istanbul, Turkey Sep 2020 – June 2026
B.S. in Electrical & Electronics Engineering (Track in Artificial Intelligence)
B.S. in Computer Engineering (Double Major)
Cumulative GPA: 3.72, Area GPA: 3.81 (100% merit scholarship)

Publications

- [A Systematic Study of In-the-Wild Model Merging for Large Language Models](#) [🔗](#)
Transactions on Machine Learning Research (TMLR), 13, March 2026.
Authors: Oğuz Kağan Hitit, Leander Gırrbach, Zeynep Akata
- [Ion Count-Aided Microscopy for Quantitative, Shot Noise-Mitigated Secondary Electron Imaging](#) [🔗](#)
Microscopy and Microanalysis, Vol. 30, Suppl. 1, July 2024.
Authors: Akshay Agarwal, Leila Kasaei, Xinglin He, Ruangrawee Kitichotkul, **Oguz Kagan Hitit**, J. Albert Schultz, Leonard C. Feldman, Vivek Goyal
- [Shot Noise-Mitigated Secondary Electron Imaging with Ion Count-Aided Microscopy](#) [🔗](#)
Proceedings of the National Academy of Sciences, Vol. 121, No. 31, July 2024.
Authors: Akshay Agarwal, Leila Kasaei, Xinglin He, Ruangrawee Kitichotkul, **Oguz Kagan Hitit**, Minxu Peng, J. Albert Schultz, Leonard C. Feldman, Vivek Goyal
- [Fourier-ring Correlation Resolution for Time-resolved Measurement in Charged Particle Microscopy](#) [🔗](#)
Microscopy and Microanalysis, Vol. 29, Suppl. 1, Aug. 2023.
Authors: **Oguz Kagan Hitit**, Akshay Agarwal, Vivek Goyal

Research Experience

Graduate Student Researcher, UCLA Ozcan Lab [🔗](#) Sep 2026 – Present
Advisor: [Prof. Aydoğan Özcan](#) [🔗](#)

- Developing a video analysis framework that combines temporal action segmentation and procedural error detection to quantify procedural reliability in laboratory evaluations of a Lyme disease point-of-care sensor, using recordings of technicians performing the tests.

Researcher, Helmholtz Munich (TUM) Explainable ML Group [🔗](#) July 2025 – March 2026
Advisor: [Prof. Zeynep Akata](#) [🔗](#)

- Benchmarked conventional and subspace-based LLM merging methods across multiple evaluation tasks to assess cross-task generalization.
- Analyzed the theoretical and empirical limits of constructive interference in model merging; first author of a TMLR publication (2026).

Researcher, KUIS AI Center [🔗](#) July 2025 – Present
Advisor: [Assoc. Prof. Aykut Erdem](#) [🔗](#)

- Developing an image-to-LoRA framework for one-shot content- and style-conditioned image generation using separate content and style hypernetworks.
- Investigating the composition of image-conditioned LoRA adapters to enable content-guided generation, style-guided generation, and reference-based stylization without per-reference fine-tuning.

UG Researcher Assistant, Koç University GGLab [🔗](#) June - Aug 2024
Advisor: [Asst. Prof. Gözde Gül Şahin](#) [🔗](#)

- Integrated lightweight NLI models with Dialogue State Tracking (DST) on MultiWOZ 2.4 to enhance conversational AI.
- Implemented BIO-tagging-based slot extraction and evaluated end-to-end improvements.

UG Researcher Assistant, Boston University STIR Group 

Sep 2022 – Aug 2024

Advisor: **Prof. Vivek Goyal** 

- Developed probabilistic algorithms to mitigate noise in particle-beam microscopy images, improving data reliability.
- Analyzed and optimized resolution-enhancement techniques for higher-quality imaging.
- Co-authored three publications, presented a poster at Microscopy & Microanalysis (M&M) 2023.
- Investigated translation of odor molecules into neural activity patterns in the olfactory circuit.
- Conducted real-world scattering experiments to improve mathematical modeling of odor conduction.

Work Experience

Co-Founder, Leagle , Istanbul, Turkey

April 2024 – Present

- Co-founded and led the development of Leagle, an AI-powered legal research and drafting platform serving 1,000+ daily users.
- Designed and deployed LLM-powered agentic workflows for legal research, document analysis, and drafting, reducing repetitive manual work for legal professionals.
- Led the end-to-end development of Leagle's NLP infrastructure and web platform, from retrieval and model orchestration to backend APIs and production deployment.

Software Engineer, Valensas , Istanbul, Turkey

July – Dec 2024

- Designed and implemented backend APIs for enterprise fintech solutions, including the *Loyalty* and *Paranoi* platforms for Turkish Airlines.
- Integrated third-party APIs and services into core systems to ensure seamless interoperability.
- Built tailored Docker images for deploying GraalVM-based applications across diverse production environments.

Software Engineering Intern, Koç University IT Department, Istanbul, Turkey

Jan – July 2024

- Designed and developed the Flutter-based *KUGate* app to streamline campus access and security workflows.

Software Engineering Intern, ASELSAN , Ankara, Turkey

Aug – Sep 2023

- Developed a Unity-based infrared gesture recognition system for potential integration into helmet avionics.
- Implemented a UDP-based data transmission protocol enabling low-latency communication for remote pilot training simulations.

Honors

- **62nd** nationwide in Turkey's 2020 University Entrance Exam (*YKS*); among > 2,000,000 candidates.
- Koç University *Dean's Honor Roll*, 2022–2026.
- **3rd** place, TUBITAK Undergraduate Research Projects Competition (2021), Defense, Space & Aeronautics category.

Skills

- **Languages & Libraries:** Python, Java, Kotlin, C, SQL; PyTorch, CUDA; Django, Spring Boot, React, Flutter
- **Systems & Tools:** PostgreSQL, Microsoft Azure, Git, Linux; Docker/Kubernetes (k3s), Slurm; Firebase
- **Research Areas:** Model merging theory; generative style transfer in T2I models; reinforcement learning in natural language processing applications; computational imaging & microscopy.